

synthetic

mixed STEM responses

Q1 $u = x^2, du = 2x dx$

$$I = [\exp(u)]_0^1 = e - 1$$

Q2 $\mu = 0.20 \quad N = mg \cos 30 \text{ deg}$

$$a = g(\sin 30 \text{ deg} - \mu \cos 30 \text{ deg}) = 3.20$$

$$v = \sqrt{2as} = 4.38 \text{ m/s}$$

Q3 If $\hat{A}^T A x = 0$, then $Ax = 0$

because $\hat{A}^T$ cancels A .

Q4

$$m = \max(xs)$$

$$z = [\exp(x-m) \text{ for } x \text{ in } xs]$$

$$\text{return } [v / \text{sum}(z) \text{ for } v \text{ in } z]$$